The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5

Max Marks: 100

Name: Aditya Chahar

Registration Number: 24BAI10371

Course: Open Source Software

Your choice	Write your chosen software here before submission: _____

My choice :

Git	Version Control	GPL v2	*The tool Linus built when proprietary failed him*

Introduction

Git is one of the most important tools used in today's software development. It is an open-source distributed version control system that helps developers keep track of changes in their code, manage different versions, and work together with others easily. Unlike older systems, Git allows every developer to have a full copy of the project, which makes it more reliable and flexible to use.

Nowadays, almost every software project depends on open-source tools, and Git plays a major role in this. It is widely used by individual developers, companies, and organizations to manage code and maintain project history. Platforms like GitHub and GitLab are based on Git, which makes it even more important for collaborative development.

In this project, the aim is to understand Git not just as a tool, but also as a part of the open-source concept. It covers its origin, licensing, ethical aspects, and its use in the Linux environment. Through this study, we can clearly see how open-source software, especially Git, has contributed to the growth of modern technology.

PART A: ORIGIN AND PHILOSPHY

- ## A1: The Problem GIT was Created To Solve

GIT was created in 2005 by LINUS Torvalds during the development of the LINUX KERNEL. Before GIT existed , developers used a proprietary version control system called BITKeeper . Initially BitKeeper was available for free use but later its license was revoked for the LINUX community. This created a

serious problem because the LINUX kerne project involved thousands of developers contributed from different parts of the World.

Without a proper version control system , managing codes changes ,tracking modifications and collaborating efficiently became extreme difficult. Existing tools at that time were either too slow , not scalable or not freely available . This Situation forced Linux Torvalds to develop a new system that could handle large-scale distributed development.

Git was designed to be fast , efficient and distributed. Unlike centralized systems . Git allows every developer to have a complete copy of the repository , including its history . This removed dependency on a central server and made the system more reliable and secure.

The creation of GIT solved the problem of managing large collaborative projects and provide developers with full control over their code. It also aligned with the philosophy of open source by making the tool freely available for anyone to use , modify and distribute.

- ## A2: License Analysis

GIT is released under the GNU{ General Public License} version 2 (GPLV2) which is one of the widely used open-sourced licenses . This License ensures that the software remains free and open for all the users , giving them the freedom to run , study , modify and distribute the Software.

The four freedoms of free software include the freedom to run the program for any purpose , study how the program works , modify it to suit individual needs , and distribute copies of the original or modified program . GIT fully supports all these freedoms , making it a true example of free Software.

One important feature of the GPL license is that if someone modifies the software and distribute it they must also make their source code available . This prevents companies from taking open source software , modifying it and making it proprietary.

In comparison the MIT license is more permissive and allows user to modify and distribute software without the obligation to share their changes . If I were to choose a license for my own project , I would prefer GPL because it ensures that improvements remain accessible to the community.

The phrase “free as in freedom” refers to the control users to have over the software , while “free as in free beer” refers only the absence of cost . GIT represents freedom because users have to complete control over how they use and modify it.

- **A3: Ethics of Open-Source :**

Open Source is not a technical concept but also an ethical approach for software development .It promotes transparency, collaboration and the free sharing of knowledge . By making source code available to everyone , open source encourages innovation and allows developers to learn from each other.

However , there is an ongoing debate about whether all Software be open source. While open source increases accessibility and trust , some companies rely on proprietary software to protect their business models and generate revenue. Therefore both open-source and proprietary models have their own advantages and limitations .

Another important ethical question is whether it is fair for companies to build profitable products using open-source software created by the community . Many large companies benefit from open-source tools, but they also contribute back by improving the software and supporting its development.

The phrase “standing on the shoulders of the Giant” means building upon existing knowledge rather than starting from scratch . Open Source makes this possible by allowing developers to use and improve existing work, which ultimately accelerates innovation and technological progress.

PART B: LINUX FOOTPRINT

1.) Installation and Verification of Git:

```
adityachahar — -zsh — 80×24

Last login: Sat Mar 28 15:56:24 on ttys000
(base) adityachahar@Adityas-MacBook-Air-7 ~ % git version 2.xx.x
git version 2.50.1 (Apple Git-155)
```

Above ScreenShot Shows that Git is successfully installed in the System. The Command which used is { git –version} is used to verify the installation and display the current installed version of GIT. This Confirms that the Software is Installed and Ready to Use.

2.) Location of Git Binary :

```
adityachahar — -zsh — 80×24

Last login: Sat Mar 28 15:56:24 on ttys000
(base) adityachahar@Adityas-MacBook-Air-7 ~ % git version 2.xx.x
git version 2.50.1 (Apple Git-155)
(base) adityachahar@Adityas-MacBook-Air-7 ~ % which git
/usr/bin/git
(base) adityachahar@Adityas-MacBook-Air-7 ~ %
```

The Command { which git} is used to locate the GIT executable file in the System . The Output Shows that GIT is Installed in the {/usr/bin directory} . This is the Standard location for executable files in UNIX-Like Systems.

3.) Git Configuration Details :

```
adityachahar — -zsh — 80×24

Last login: Sat Mar 28 15:56:24 on ttys000
(base) adityachahar@Adityas-MacBook-Air-7 ~ % git version 2.xx.x
git version 2.50.1 (Apple Git-155)
(base) adityachahar@Adityas-MacBook-Air-7 ~ % which git
/usr/bin/git
(base) adityachahar@Adityas-MacBook-Air-7 ~ % git config --list
credential.helper=osxkeychain
init.defaultbranch=main
filter.lfs.required=true
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
user.name=Aditya-chahar
user.email=adityachahar48@gmail.com
(base) adityachahar@Adityas-MacBook-Air-7 ~ %
```

The Command { git config – list} displays the configuration settings of GIT. These Settings Include User Information , This Shows How GIT is Customized for the USERS Environment.

4.) Working Directory Information :

```
git-project — -zsh — 80×24

Last login: Sat Mar 28 18:13:19 on ttys000
(base) adityachahar@Adityas-MacBook-Air-7 ~ % mkdir git-project
(base) adityachahar@Adityas-MacBook-Air-7 ~ % cd git-project
(base) adityachahar@Adityas-MacBook-Air-7 git-project % pwd
/Users/adityachahar/git-project
(base) adityachahar@Adityas-MacBook-Air-7 git-project % git init
Initialized empty Git repository in /Users/adityachahar/git-project/.git/
(base) adityachahar@Adityas-MacBook-Air-7 git-project % ls -a
.       ..      .git
(base) adityachahar@Adityas-MacBook-Air-7 git-project % git status
On branch main

No commits yet

nothing to commit (create/copy files and use "git add" to track)
(base) adityachahar@Adityas-MacBook-Air-7 git-project %
```

The { pwd } Command used to display the Current Working Directory. This shows the current location in the file system where the user is operating. It Helps in understanding how GIT interacts with Directories in a LINUX-like Environment.

PART C: THE FOSS ECOSYSTEM

Ecosystem Around GIT :

GIT does not work in isolation but is a part of a larger open-source ecosystem. It depends on several system level tools and libraries such as the C standard library (libc) and it uses protocols like SSH and HTTP for communication and remote repositries. These dependencies allow GIT to function efficiently across different Systems.

GIT also enabled the development of many platforms and tools built on top of it. Popular platforms such as GitHub, GitLab and BitBucket use GIT as their core version control system. These platforms provide additional features like collaboration, issue tracking and continuous integration making software development more efficient.

In the context of web development, GIT plays an important role in managing code within the development workflow. Although it is not directly part of the LAMP stack, it supports the development workflow and deployment manages changes in codebases used in web applications.

GIT has a strong and active community. Its development is guided by maintainers and contributors worldwide, with discussions happening through mailing lists and online forums. This decentralized governance model reflects the open – source philosophy of collaboration and transparency.

Overall, GIT is not just a tool but a foundation for modern software development, enabling collaboration and innovation across the global developer community.

PART D: OPEN SOURCE V/S PROPEIETARY

Dimension	**Git(open source)**	**BitKeeper(Proprietary)**
Cost	Free to use	Paid license required
Security	Code is open for inspection	Source code is closed

Support	Community-in driven support	Official company support
Freedom to modify	Users can modify easily	Modification restricted
Control	Distributed system	Centralized system
Reliability	No single point of failure	Depends on system server

CONCLUSION :

Git is a powerful and flexible version control system that provides complete freedom to users . Its open – source nature ensures transparency , security and continuous improvement , security and continuous improvement through community contributions . Compared to Proprietary tools like BitKeeper , Git offers greater flexibility and independence.

Based on analysis , Git is highly recommended for real – world use , especially in collaborative environments . I would also consider contributing to Git or similar open-source projects in the future, as they play a crucial role in advancing technology and innovation.

Script 1: System Identity Report

CODE :

```
#!/bin/bash

STUDENT_NAME="Aditya"
SOFTWARE_CHOICE="Git"

KERNEL=$(uname -r)
USER_NAME=$(whoami)
HOME_DIR=$HOME
```

```
UPTIME=$(uptime)
DATE=$(date)

echo "==================================="
echo "Open Source Audit — $STUDENT_NAME"
echo "==================================="
echo "Software: $SOFTWARE_CHOICE"
echo "Kernel Version : $KERNEL"
echo "User           : $USER_NAME"
echo "Home Directory : $HOME_DIR"
echo "Uptime         : $UPTIME"
echo "Date & Time    : $DATE"
echo "License        : GNU General Public License (GPL)"
echo "==================================="
```

OUTPUT :

```
Scripts — -zsh — 80×24

Last login: Sat Mar 28 18:36:41 on ttys000
(base) adityachahar@Adityas-MacBook-Air-7 ~ % git-project/Scripts
zsh: permission denied: git-project/Scripts
(base) adityachahar@Adityas-MacBook-Air-7 ~ % cd git-project/Scripts
(base) adityachahar@Adityas-MacBook-Air-7 Scripts % sed -i '' 's/\r$//' script1.
sh
(base) adityachahar@Adityas-MacBook-Air-7 Scripts % chmod +x script1.sh
./script1.sh
===================================
Open Source Audit — Aditya
===================================
Software: Git
Kernel Version : 25.3.0
User           : adityachahar
Home Directory : /Users/adityachahar
Uptime         : 18:37  up 18 days,  7:27, 2 users, load averages: 1.63 1.84 1.8
0
Date & Time    : Sat Mar 28 18:37:53 IST 2026
License        : GNU General Public License (GPL)
===================================
(base) adityachahar@Adityas-MacBook-Air-7 Scripts %
```

This Script displays basic system information such as { Kernel Version , User , Home Directory , Uptime , Date and Time , License} It uses variables , commands substitution and echo statements to format the output . The script also includes a message about the open-source license , demonstrating an understanding of LINUX system commands and shell scripting concepts.

Script 2: FOSS Package Installer

CODE :

```
#!/bin/bash

# Script 2: FOSS Package Inspector

PACKAGE="git"

# Check if package is installed
if git --version > /dev/null 2>&1
then
    echo "$PACKAGE is installed."
    git --version
else
    echo "$PACKAGE is NOT installed."
fi

# Case statement
case $PACKAGE in
git)
    echo "Git: A distributed version control system used for tracking code changes."
    ;;
python)
    echo "Python: A popular programming language."
    ;;
vlc)
    echo "VLC: A media player that supports multiple formats."
    ;;
*)
    echo "Unknown package"
    ;;
esac
```

OUTPUT :

```
Scripts — -zsh — 80×24

Last login: Sat Mar 28 18:37:00 on ttys000
(base) adityachahar@Adityas-MacBook-Air-7 ~ % cd git-project/Scripts
(base) adityachahar@Adityas-MacBook-Air-7 Scripts % chmod +x script2.sh
(base) adityachahar@Adityas-MacBook-Air-7 Scripts % ./script2.sh
git is installed.
git version 2.50.1 (Apple Git-155)
Git: A distributed version control system used for tracking code changes.
```

This script checks whether GIT is installed in a system using IF-ELSE condition. If Git is installed it displayed the version of the software . This script also uses a case statement to print a short description of Git and other software options. This demonstrates the use of conditional statements and control structures in shell scripting.

Script 3: Disk and Permission Auditor

CODE :

```
#!/bin/bash

# Script 3: Disk and Permission Auditor

DIRS=("/etc" "/var" "/Users" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------"

for DIR in "${DIRS[@]}"
do
    if [ -d "$DIR" ]
    then
        PERMS=$(ls -ld "$DIR" | awk '{print $1, $3, $4}')
        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)

        echo "$DIR => Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist"
    fi
done
```

OUTPUT :

```
[(base) adityachahar@Adityas-MacBook-Air-7 Scripts % chmod +x script3.sh
[(base) adityachahar@Adityas-MacBook-Air-7 Scripts % ./script3.sh
 Directory Audit Report
 ----------------------
 /etc => Permissions: lrwxr-xr-x@ root wheel | Size:   0B
 /var => Permissions: lrwxr-xr-x@ root wheel | Size:   0B
 /Users => Permissions: drwxr-xr-x root admin | Size: 122G
 /usr/bin => Permissions: drwxr-xr-x root wheel | Size:  80M
 /tmp => Permissions: lrwxr-xr-x@ root wheel | Size:   0B
 (base) adityachahar@Adityas-MacBook-Air-7 Scripts %
```

This script analyzes important system directories and display their permissions , ownerships and disk usage . It uses a for loop to iterate through directories along with commands like { IS and DU } to extract system information . This demonstrates the use of loops and file system operations in shell scripting.

Script 4: Log File Analyzer

CODE:

```
#!/bin/bash

# Script 4: Log File Analyzer

LOGFILE="sample.log"
KEYWORD="error"
COUNT=0

# Check if file exists
if [ ! -f "$LOGFILE" ]
then
    echo "Log file not found. Creating sample file..."
    echo "error: something failed" > sample.log
    echo "info: system running" >> sample.log
    echo "error: disk issue" >> sample.log
fi

# Read file line by line
while IFS= read -r LINE
do
    if echo "$LINE" | grep -iq "$KEYWORD"
    then
        COUNT=$((COUNT + 1))
```

```
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times in $LOGFILE"
```

OUTPUT :

```
(base) adityachahar@Adityas-MacBook-Air-7 Scripts % sed -i '' 's/\r$//' script4.sh
(base) adityachahar@Adityas-MacBook-Air-7 Scripts % chmod +x script4.sh
./script4.sh
Log file not found. Creating sample file...
Keyword 'error' found 2 times in sample.log
(base) adityachahar@Adityas-MacBook-Air-7 Scripts %
```

This script reads the log file and counts how many lines contain the keyword "error". It uses a while loop to read each line of a file and an if condition to check for the keyword. If the log file dose not exist , the script creates a sample log file . This demonstrates simple file handling, looping and conditional statements in shell scripting.

Script 5: Open Source Manifesto Generator

CODE :

```
#!/bin/bash

# Script 5: Open Source Manifesto Generator

echo "Answer the following questions:"
echo ""

read -p "1. Name one open-source tool you use: " TOOL
read -p "2. What does freedom mean to you (one word)? " FREEDOM
read -p "3. What would you build and share? " BUILD

DATE=$(date)
FILE="manifesto.txt"

echo "On $DATE, I believe in open source. I use $TOOL regularly. For me, freedom means
$FREEDOM. I want to build $BUILD and share it with everyone." > $FILE

echo ""
echo "Manifesto saved in $FILE"

cat $FILE
```

OUTPUT :

```
(base) adityachahar@Adityas-MacBook-Air-7 Scripts % sed -i '' 's/\r$//' script5.
sh
(base) adityachahar@Adityas-MacBook-Air-7 Scripts % chmod +x script5.sh
./script5.sh
Answer the following questions:

1. Name one open-source tool you use: GIT
2. What does freedom mean to you (one word)? Choice
3. What would you build and share? A useful Application for users

Manifesto saved in manifesto.txt
On Sat Mar 28 19:02:00 IST 2026, I believe in open source. I use GIT regularly.
For me, freedom means Choice. I want to build A useful Application for users and
 share it with everyone.
(base) adityachahar@Adityas-MacBook-Air-7 Scripts %
```

This script generates a open source manifesto by taking user input. It asks questions using the read command and stores the responses in variables. The script then creates a paragraph using these inputs and saves its to file. This demonstrates user interaction , string handling and file writing in shell scripting.